

A Convolution Interface for the Literal Prime Shell and the Divisibility–Mask Defect

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September 2, 2026

Abstract

We isolate the two mechanisms in the literal deleted–diagonal prime–shell matrix used in the current twin–prime route. The residue kernel is a translation–invariant convolution on $\mathbb{Z}$, whereas the two endpoint divisibility masks are coordinate projections. We prove an exact convolution–plus–defect factorisation, the Fourier multiplier formula for the unmasked convolution on $\ell^2(\mathbb{Z})$, and a finite triangle envelope which retains the mask defect. A guarded audit on two disjoint origins, three source counts, four shells, two kernel exponents, and four sign laws contains 192 rows. The ideal matrix is translation invariant in all 96 parameter groups, but the defect-to-ideal spectral ratio reaches 0.467075645603 and exceeds $1/4$ in 93 rows. The result is a precise interface and a finite obstruction to discarding the masks; it is not an arithmetic L^2 estimate, a power saving, or a twin-prime theorem.

1 Question and scope

The recent finite route has repeatedly encountered a distinction between a signed source calculation and the physical output operator. Here we return to the unresolved source-uniform L^2 question at the operator level. The question is deliberately narrow: can the literal matrix be compared with a translation-invariant object without erasing the divisibility information?

Throughout, $Q < p \leq 2Q$ ranges over primes, $H = 66$, and $s \in \{1, 2\}$. For $d \in \mathbb{Z}$ define

$$k_p(d) = \begin{cases} 0, & d = 0, \\ p \frac{H^{2s}}{(H^2 + d^2)^s} \left(\mathbf{1}_{p|d} - \frac{1}{p-1} \right), & d \neq 0. \end{cases} \quad (1)$$

For a sign law $e = (e_p)$ let $K_e = \sum_{Q < p \leq 2Q} e_p K_p$, where $(K_p f)(u) = \sum_d k_p(d) f(u - d)$. The physical finite matrix on an interval I has entries

$$A_I(u, t) = \sum_p e_p \mathbf{1}_{u \neq t} \mathbf{1}_{p \nmid u} \mathbf{1}_{p \nmid t} p \frac{H^{2s}}{(H^2 + (u - t)^2)^s} \left(\mathbf{1}_{p|u-t} - \frac{1}{p-1} \right). \quad (2)$$

No source vector is needed for this interface audit. In particular, the finite table below must not be read as evidence for a twin-prime density.

2 Exact factorisation

Let P_p be multiplication by $\mathbf{1}_{p \nmid n}$ on sequences, and let E_I extend a vector on I by zero while R_I restricts a sequence to I .

Proposition 1 (mask factorisation). *With K_p defined by (1), the physical operator is*

$$A_I = \sum_p e_p R_I P_p K_p P_p E_I. \quad (3)$$

If $T_I = R_I K_e E_I$ and $D_I = A_I - T_I$, then

$$D_I = \sum_p e_p R_I ((P_p - I) K_p P_p + K_p (P_p - I)) E_I. \quad (4)$$

Proof. The two projections in (3) remove exactly the entries whose endpoints are divisible by p , and $k_p(0) = 0$ removes the diagonal. This gives (2) entry by entry. For one prime, $(P - I)KP + K(P - I) = PKP - K$; summing and applying restriction and extension gives (4). All sums over the shell are finite. $\square$

The decomposition is more informative than a generic Frobenius envelope: T_I is the part to which Fourier analysis applies, while D_I records every position-sensitive mask contribution.

3 The unmasked Fourier interface

Since $|k_p(d)| \ll_{p,H,s} |d|^{-2s}$ for $d \neq 0$, $k_e \in \ell^1(\mathbb{Z})$. Its Fourier series

$$\widehat{k}_e(\theta) = \sum_{d \in \mathbb{Z}} k_e(d) e^{-id\theta}, \quad -\pi \leq \theta \leq \pi, \quad (5)$$

is continuous and periodic.

Theorem 1 (Fourier and compression bounds). *For the explicitly unmasked operator K_e ,*

$$\|K_e\|_{\ell^2(\mathbb{Z}) \rightarrow \ell^2(\mathbb{Z})} = \operatorname{ess\,sup}_{\theta \in [-\pi, \pi]} |\widehat{k}_e(\theta)| \leq \sum_{d \in \mathbb{Z}} |k_e(d)|. \quad (6)$$

For every finite interval I ,

$$\|A_I\| \leq \|T_I\| + \|D_I\| \leq \|K_e\| + \|D_I\|_F. \quad (7)$$

Proof. The Fourier transform on $\ell^2(\mathbb{Z})$ is unitary and sends convolution by k_e to multiplication by (5); the norm of a multiplication operator is the essential supremum. Young’s inequality gives the second inequality in (6). Since R_I and E_I have norm one, $\|T_I\| \leq \|K_e\|$. Apply the triangle inequality to $A_I = T_I + D_I$ and then $\|D_I\| \leq \|D_I\|_F$. $\square$

For $R \geq 1$, the tail in the Young envelope has the explicit bound

$$\sum_{|d| > R} |k_e(d)| \leq \frac{2H^{2s} \sum_{Q < p \leq 2Q} p}{(2s - 1)R^{2s-1}}. \quad (8)$$

Indeed, the centered residue factor has absolute value at most one and $\sum_{d > R} d^{-2s} \leq R^{1-2s}/(2s - 1)$. Equation (8) is a sign-free baseline; it supplies no cancellation.

4 Finite audit

The protocol uses origins 40097 and 48097, source counts $M \in \{256, 512, 1024\}$, $Q \in \{24, 36, 54, 80\}$, $s \in \{1, 2\}$, and the four sign laws ‘all-plus’, alternating shell index, the mod-4 character, and the half split. Thus $2 \cdot 3 \cdot 4 \cdot 2 \cdot 4 = 192$ physical/ideal/defect rows. The Young finite sum uses $R = 65536$ and the analytic tail in (8).

For each row we compute the symmetric spectral norms of A_I , T_I , and D_I , and the Frobenius norm of D_I . The certificate also checks

$$\|A_I\| \leq \|K_e\| + \|D_I\|_F. \quad (9)$$

For each fixed (M, Q, s, e) , translating the interval changes neither the differences nor the ideal matrix. There are 96 such two-origin checks.

The ideal translation identity is exact at the displayed operator level; the 96/96 label records its independent floating-point replay. The defect ratio is not a monotone function of shell size or interval count: it is a diagnostic of the masks and their placement, not a candidate exponent.

5 Adversarial checks and claim boundary

The independent checker rebuilds the shells and matrices in reverse prime order, recomputes all spectral quantities, and verifies the canonical JSON payload. A rational six-point anchor with $Q = 4$ and shell $\{5, 7\}$ checks the identity $A_I = T_I + D_I$ and symmetry without floating point. The stress suite attacks a sign-flipped defect, a reintroduced diagonal, random projection contractions, and the analytic tail scale. Normal and optimized executions agree byte-for-byte at the reporting layer and produce empty standard error.

The result supports the following narrow ledger:

Table 1: Summary of the 192-row finite spectral audit.

quantity	certified finite readout
physical rows	192
ideal translation checks	96/96
combined-envelope checks	192/192
minimum $\ D_I\ / \ T_I\ $	0.0312337689685
maximum $\ D_I\ / \ T_I\ $	0.467075645603
rows with $\ D_I\ / \ T_I\ > 1/4$	93/192
range of $\ A_I\ / \ T_I\ $	0.938011283061–1.13920880401
range of combined occupancy	0.121634514969–0.682234184555

statement	status
mask factorisation and defect identity	proved exact finite
unmasked Fourier multiplier interface	proved exact conditional
Young envelope and interval compression	proved exact
192-row spectral replay	numerically certified finite
discarding masks on this declared panel	refuted scoped
source-uniform arithmetic L^2	open
fixed-power credit / Route-B Gate B	0 / open
twin-prime conclusion	none

Remark 1. *The finite obstruction does not say that every possible mask estimate fails, nor that the defect has a positive asymptotic lower bound. It says only that the current finite physical object is not faithfully represented by its unmasked convolution on the tested panel. Any future arithmetic L^2 argument must either control D_I with its position dependence or prove cancellation for the masked sum directly.*

6 Conclusion and next question

The main route contribution is a typed interface: Fourier analysis is available for a precisely identified unmasked component, and the missing piece is an explicit operator defect rather than an implicit normalization error. The finite audit shows that this defect is sometimes substantial, with 93 of 192 rows above the quarter threshold. We therefore freeze the shortcut of dropping the masks and return to the position-aware arithmetic L^2 problem. No arithmetic advance is claimed in this release.